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SEAT No. _____

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SARDAR PATEL UNIVERSITY
M.Sc.IT (Integrated) EXAMINATION, SEMESTER III
Thursday, 25th October, 2018
10:00 a.m. to 1:00 p.m.

PS03CHT02 – Advanced Data and File Structures

Maximum Marks: 70

- Q-1 Multiple Choice Question.[Each Question carries one Mark] [10]**
- 1) The line drawn from one node to another node is called _____.
 (a) Degree. (b) Leaf. (c) Edge. (d) Height
 - 2) The nodes which have the same parent are called _____.
 (a) Degree. (b) Leaf. (c) Height. (d) Siblings
 - 3) If the range of index varies from L..... U then size of the array is _____.
 (a) $U - L + 1$. (b) $L - U + 1$. (c) $U / L - 1$. (d) $L / U - 1$.
 - 4) Minimum number of nodes possible in a binary tree of height h is _____.
 (a) $h + 1$ (b) $h - 1$ (c) h (d) 2^h
 - 5) Preorder traversal of a binary tree can be visited _____ manner.
 (a) Left sub-tree, Root, Right sub-tree.
 (b) Root, Left sub-tree, Right sub-tree.
 (c) Root, Right sub-tree, Left sub-tree.
 (d) Left sub-tree, Right sub-tree, Root
 - 6) If there are more than one edges between the same pair of vertices, then they are known as _____.
 (a) Different edges. (b) Same edges. (c) Self loop. (d) Parallel edges
 - 7) The best case performance of the bubble sort is _____ comparisons.
 (a) $n + 1$ (b) n (c) $n - 1$ (d) $n / 2$
 - 8) K-way merging is known as _____.
 (a) Simple merge (b) Multiple merging (c) Selection sort (d) Binary merging
 - 9) _____ processing is the accessing of records one after the other, in ascending order by a key or index item of the record.
 (a) Serial (b) Sequential (c) Direct (d) None of these
 - 10) The open addressing and chaining are two classes of _____ techniques.
 (a) Collision Resolution (b) Hashing
 (c) Blocking (d) None of these

①

(PTO)

Q-2 Give Answers for the following:(Any ten)

1. Define Array. What is the upper bound and lower bound of an array?
2. Write the formula for address calculation of 2-D array element and explain it.
3. Define : Node and Branch
4. Write the sequence of steps for Inorder tree traversal.
5. Draw the Binary Tree for $(A-B)+c*(E/F)$
6. What is loop and cycle of a Graph?
7. Difference between Searching and sorting.
8. List the applications of sorting.
9. Define Binary Search.
10. Explain in brief prime area.
11. Explain Serial processing in brief.
12. Define File organization and list out different types of file organization.

- Q -3** A) Explain 2-D array with declaration and initialization [5]
 B) Explain Sparse Matrix in detail. [5]

OR

- Q -3** A) Explain 1-D array with declaration and initialization. [5]
 B) Assume that the base address of the two dimensional array $a[10][10]$ is 250, each element requires 2 byte (word). Find the address of the element $a[2][3]$ using [5]
 (a) Row-major order
 (b) Column-major order

- Q -4** A) What are the types of traversal of Binary tree? Explain any two with an example. [5]
 B) Explain deletion of a node from binary tree (all 3 conditions) with example. [5]

OR

- Q -4** A) Define Graph. Explain Directed graph, Undirected graph, connected graph, weekly connected graph with an example. [5]
 B) Draw the binary tree for following expressions: [5]

In order	n1	n2	n3	n4	n5	n6	n7	n8	n9
Post order	n1	n3	n5	n4	n2	n8	n7	n9	n6

- Q -5** What is searching? List and explain searching techniques with algorithms. [10]

OR

- Q -5** What is sorting? List sorting techniques and explain any one of them with algorithm. [10]

- Q -6** A) Write a short note on Multiple buffering. [5]
 B) Explain the structure of index sequential file supported by IBM. [5]

OR

- Q -6** A) Write a short note on Single buffering. [5]
 B) Explain the structure of index sequential file supported by CDC. [5]